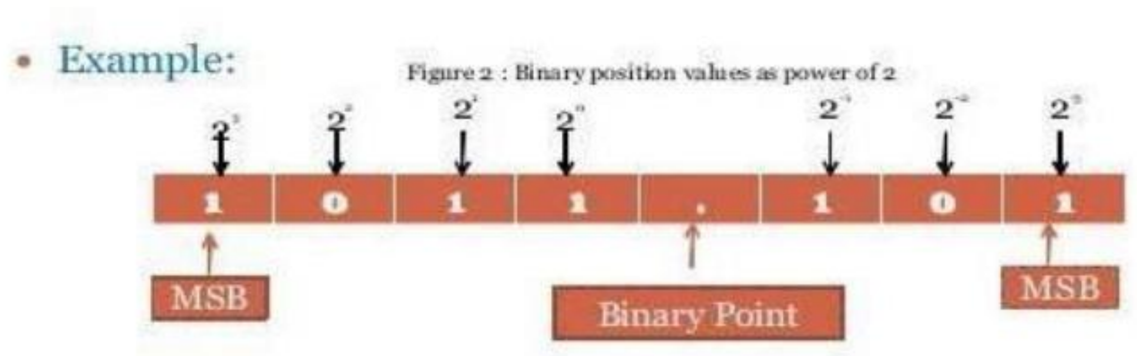
BINARY CONVERSIONS

Binary number system:

The binary number has a radix of 2. As r = 2, only two digits are needed, and these are 0 and 1. In binary system weight is expressed as power of 2.



The left most bit, which has the greatest weight is called the Most Significant Bit (MSB). And the right most bit which has the least weight is called Least Significant Bit (LSB).

For Ex: $1001.01_2 = [\ (1) \times 2^3\ ] + [\ (0) \times 2^2\ ] + [\ (0) \times 2^1\ ] + [\ (1) \times 2^0\ ] + [\ (0) \times 2^{-1}\ ] + [\ (1) \times 2^2\ ]$

$$1001.01_2 = [\ 1 \times 8\ ] + [\ 0 \times 4\ ] + [\ 0 \times 2\ ] + [\ 1 \times 1\ ] + [\ 0 \times 0.5\ ] + [\ 1 \times 0.25\ ]$$

$$1001.01_2 = 9.25_{10}$$

Decimal Number system

The decimal system has ten symbols: 0,1,2,3,4,5,6,7,8,9. In other words, it has a base of 10.

Octal Number System

Digital systems operate only on binary numbers. Since binary numbers are often very long, two shorthand notations, octal and hexadecimal, are used for representing large binary numbers. Octal systems use a base or radix of 8. It uses first eight digits of decimal number system. Thus it has digits from 0 to 7.

Hexa Decimal Number System

The hexadecimal numbering system has a base of 16. There are 16 symbols. The decimal digits 0 to 9 are used as the first ten digits as in the decimal system, followed by the letters A, B, C, D, E and F, which represent the values 10, 11,12,13,14 and 15 respectively.

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

BINARY CONVERIONS

1.Binary To Octal

Group bits in sets of 3 (from LSB to MSB). Convert each group to its octal digit (0–7).

The binary number:	001	010	011	000	100	101	110	111
The octal number:	1	2	3	0	4	5	6	7

2.Binary To Hexadecimal

Group bits in sets of 4. Convert each group to hex digit (0–9, A–F).

The binary number:	0001	0010	0100	1000	1001	1010	1101	1111
The hexadecimal number:	1	2	5	8	9	A	D	F

3.Binary To Decimal

- Use the sum-of-weights method:

For integer bits: each bit * 2^(position), sum them.
Bits = 0 contribute nothing.

- For fractional bits: bit * 2^(−position) (position starts from 1 for first bit after the point).

Example of Binary to Decimal Conversion:

Convert the binary number $(1101)_2$ into a decimal number.

Solution:

Given binary number = $(1101)_2$

Now, multiplying each digit from MSB to LSB with reducing the power of the base number 2.

$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

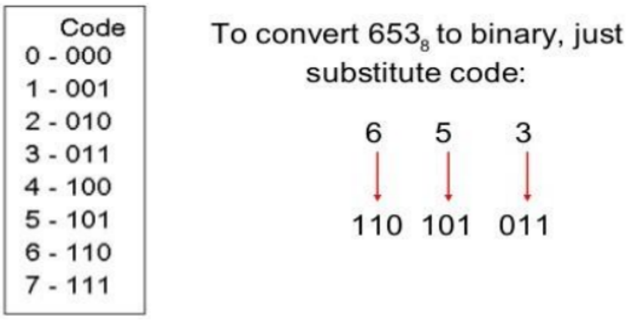
$= 8 + 4 + 0 + 1$

$= 13$

Thus, the equivalent decimal number for the given binary number $(1101)_2$ is $(13)_{10}$

4. Octal To Binary

Replace each octal digit with its 3-bit binary.



5.Hexadecimal To Binary

Replace each hex digit with its 4-bit binary.

Question : Convert $A2B_{16}$ to an equivalent binary number.

Solution: Given hexadecimal number = $A2B_{16}$

First, convert the given hexadecimal to the equivalent decimal number.

$A2B_{16} = (A \times 16^2) + (2 \times 16^1) + (B \times 16^0)$

$= (A \times 256) + (2 \times 16) + (B \times 1)$

$= (10 \times 256) + 32 + 11$

$= 2560 + 43$

= 2603(Decimal number)

Now we have to convert 2603_{10} to binary

```
2 | 2603
2 |1301 -- 1
2 | 650 -- 1
2 | 325 -- 0
2 | 162 -- 1
2 | 81  -- 0
2 | 40 -- 1
2 | 20 -- 0
2 | 10 -- 0
2 | 5 -- 0
2 | 2 -- 1
2 | 1 -- 0
2 | 0 -- 1
```

The binary number obtained is 101000101011_2

Hence, $A2B_{16} = 101000101011_2$

6.Decimal To Binary

For integer part:

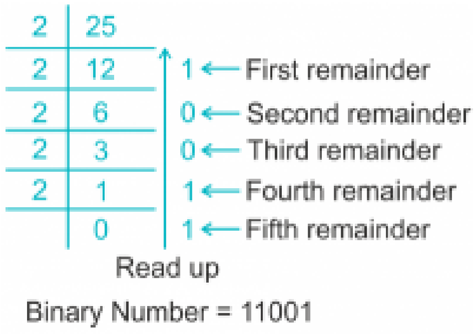
1. Repeatedly divide by 2.
2. Record the remainder (0 or 1) each time.
3. Continue until quotient becomes 0.
4. The binary representation is the remainders read bottom → top (last remainder is MSB).

   Solved Example 1: Convert the decimal number 25 into a binary number.

   Solution:

Follow the steps below to convert 25 into a binary system.

1. Divide the number 25 by 2.
2. Get the integer quotient for the next iteration.
3. Get the remainder for the binary digit.
4. Repeat the above steps until the quotient is equal to 0.



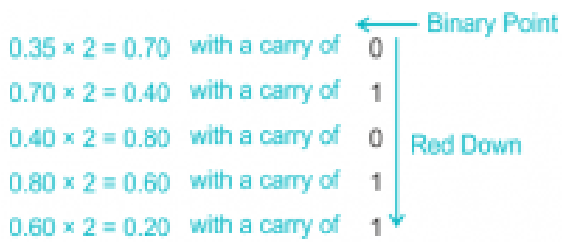
For fractional part (if any):

1. Multiply the fractional portion by 2.
2. The integer part of the result (0 or 1) is the next binary digit.
3. Keep the fractional remainder and repeat until fraction becomes 0 or you reach desired precision.

Solved Example : Convert the decimal number 0.35 into a binary number.

Solution:

Multiply by 2 and record the carry in the integral position for fractional decimal figures. When the carries are read down, the equivalent binary fraction is produced, as seen in the sample below.



Thus the fractional binary number is .01011, i.e., 0.01011.

MCQ QUESTIONS

1. What are the symbols used to represent digits in the binary number system?

a) 0,1

b) 0,1,2

c) 0 through 8

d) 1,2

2. If the decimal number is a fraction then its binary equivalent is obtained by

________ the number continuously by 2.

a) Dividing

b) Multiplying

c) Adding

d) Subtracting

3. The decimal equivalent of the binary number (1011.011)2 is ________

a) (11.375)10

b) (10.123)10

c) (11.175)10

d) (9.23)10

4. Convert binary to octal: (110110001010)2 =?

a) (5512)8

b) (6612)8

c) (4532)8

d) (6745)8

5. Octal to binary conversion: (24)8 =?

a) (111101)2

b) (010100)2

c) $(111100)_2$

d) $(101010)_2$

6. Convert the binary number $(01011.1011)_2$ into decimal:

a) $(11.6875)_{10}$

b) $(11.5874)_{10}$

c) $(10.9876)_{10}$

d) $(10.7893)_{10}$

7.In the Hexadecimal number system, the letters A-F represent digits ________

a) 10-16
b) 10-15
c) 9-15

8.To convert a decimal number into a binary number, divide the number by ________

a) 2
b) 8
c) 10

Binary conversions in Logic System Design

Links provided are

https://www.youtube.com/watch?v=E-Sg2YsvZds